Date: 26-08-2026

Function

Calculus : The field of mathematics which deals w differentiation and integration of functions and relat cocepts and applications. Calculus studied in two p mainly.

(i) Differential Calculus : Differential calculus deal with the rate of change of one quantity with anoth when the change in the latter is ultimately very small as well as other alied problems.

(ii) Integral Calculus : The study of integration as suc and its application of finding areas, volumes, centroids, equation of curves, solution of differenti equations etc.

§1.1. Introduction : The whole of calculus is based two important concepts, the concept of function. and the concept of limit and these concepts are base on the theory real numbers.

§1.2. Natural Numbers : Numbers 1, 2, 3, 4 ..... use in a ordinary counting are called natural numbers. The set of natural numbers are denoted by N.

this, $N = \{1, 2, 3, 4, \ldots\}$

The natural numbers 1,2,3,4 .... are called positive integers and the numbers $-1,-2,-3,\ldots$ are called Negative integers.

§ 1.3 Integers : The number $\ldots -3,-2,-1,0,1,2,3$ etc are called integers. The set of integers are denoted by I or Z

Thus, I or $Z = \{\ldots -3,-2,-1,0,1,2,3,\ldots\}$

$= \{0, \pm 1, \pm 2, \pm 3, \ldots\}$

clearly $N \in Z$

§ 1.4 "0" is also a integer, not a positive integer or a negative integer.

But "0" is definitely a non negative integers as well as a non positive integer.

Thus (i) set of positive integers = $\{1,2,3\ldots\}$

(ii) set of negative integers = $\{-1,-2,-3,\ldots\}$

(iii) set of non negative integers = $\{0,1,2,3\ldots\}$

(iv) set of non positive integers = $\{0,-1,-2,-3\ldots\}$

(v) if $a \neq 0$ then $\frac{0}{a} = 0$ ("0" is divisible by all non zero numbers)

vi) No number is divisible by zero.

$\frac{3}{0} = \infty$

vii) 0 is an even number because ...
by 2.

§ 1.5 Rational number : A number of the form p/q where are integer and $q \neq 0$ is called a rational number.

Example: (i) $3/2$ ii) $5/3$ (iii) 6 (as $6/1$)

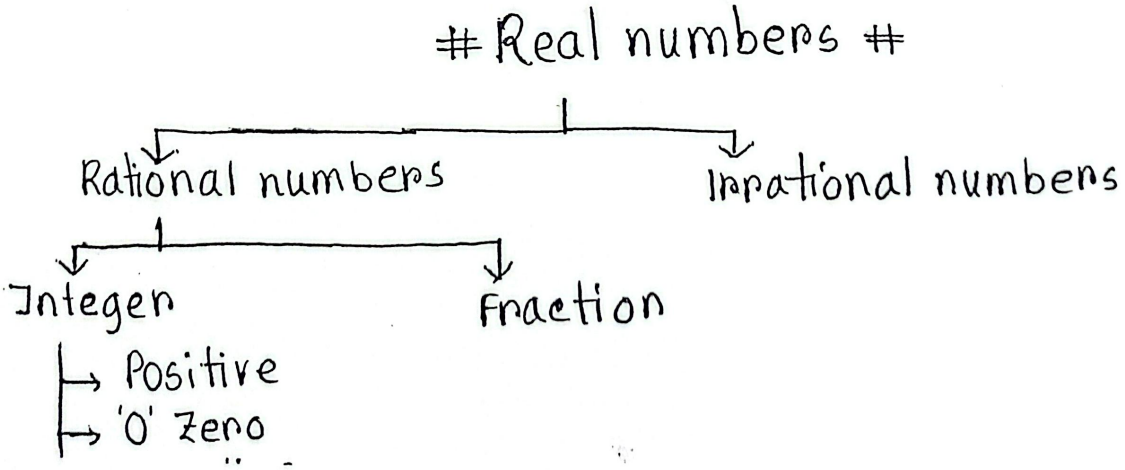
The set $Q = \{ \frac{p}{q} : p, q \in z \text{ and } q \neq 0 \}$

clearly $N \subset Z \subset Q$

§ 1.6 Irrational number :

§ 1.7 Real number : Real numbers are those whi
are either rational or irrational. The set of rea
numbers are denoted by R.

clearly $N \subset Z \subset Q \subset R$

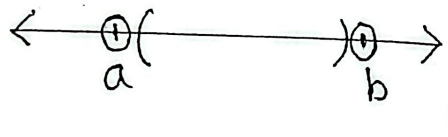


§ 1.8 Some important subset of R:

(i) Open interval : Let 'a' and 'b' be two real numbers such that $a<b$ then the set of all real numbers lying strictly between a and b is called an open interval and is denoted by (a,b) or $]a,b[$

We generally discribe as follows

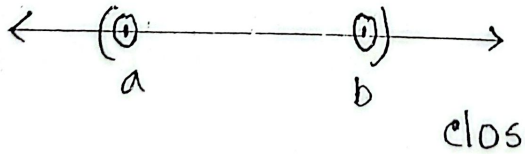
$$(a,b) = \{x : a < x < b\}$$



Open

(ii) close interval : Let 'a' and 'b' be two real numbers such that $a<b$, then the set of all real numbers lying between 'a' and 'b' including 'a' and 'b' as well is called 'a' close interval and is denoted by $[a,b]$

Thus, $[a,] = \{x : a \leq x \leq b\}$



close

§1.9 Concept of infinity : $x = \infty$ means

(i) x is larger than any real number

(ii) x is not a fixed number

$x = -\infty$ means

(i) x is smaller than any negative number however small

(ii) x is not a fixed number

Properties of ∞ and $-\infty$

(i) $\infty \pm x = \infty$, for every $x \in \mathbb{R}$

(ii) $-\infty \pm x = -\infty$, for every $x \in \mathbb{R}$

(iii) $\infty + \infty = \infty$

(iv) $\infty \cdot \infty = \infty$

(v) $(-\infty)(-\infty) = \infty$

(vi) $\infty \cdot (-\infty) = -\infty$

(vii) $(-\infty) \cdot \infty = -\infty$

(viii) $x \cdot \infty = \infty$ of $x > 0$

$x \cdot \infty = -\infty$ of $x < 0$

$x \cdot \infty = 0$ of $x = 0$

(ix) $\infty^{\infty} = \infty$

(x) $\infty - \infty =$ (meaningless)

(xi) $\frac{\infty}{\infty} =$ (meaningless)

Table-01

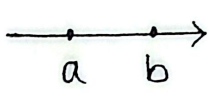
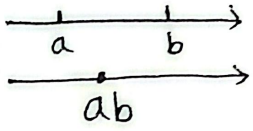
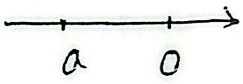
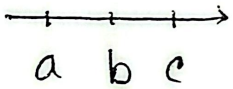
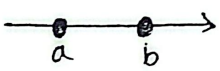
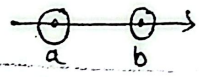
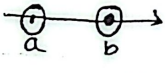
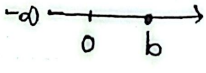
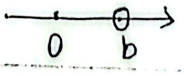
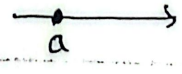
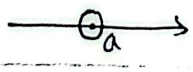
Inequality	Geometric Interpretation	Illustration
$a<b$ or $b>a$	'a' is the left of -b	
$a \leq b$ or $b \geq a$	'a' is the left of -b or coinside with b	
$0<a$ or $a>0$	'a' is the right of the Origin	
$a<0$ or $0>a$	'a' is the left of the origin	
$a<b<c$	'a' is the left of 'b' and 'b' is to the left of 'c'	

Table-02

Interval notation	set notation	Gemetric picture	classifications
$[a,b]$	$\{x: a \leq x \leq b\}$		Finite closed
(a,b)	$\{x: a<x<b\}$		finite open
$[a,b)$	$\{x: a \leq x<b\}$		half open half close
$(a,b]$	$\{x: a<x \leq b\}$		half close half open
$(-\infty,b]$	$\{x: x \leq b\}$		Infinite closed
$(-\infty,b)$	$\{x: x<b\}$		Infinite open
$[a,+\infty)$	$\{x: x \geq a\}$		Infinite closed
$(a,+\infty)$	$\{x: x>a\}$		Infinite open
$(-\infty,+\infty)$	$\{x: x$ is real number$\}$		Infinite open and close

§ 1.10 Undefined statement:

(i) Division by zero is undefined ($\frac{3}{0}$ is undefined)

(ii) Square root of a negative quantity is undefine ($\sqrt{-1}, \sqrt{-5}$)

(iii) Logarithm of a non-positive number is undefined $\{\log(-2), \log(0)\}$

§ 1.11 Absolute value: If 'a' be any real number then we have either $a=0$ or $a>0$ or $a<0$ we note further $a<0$ implies $-a>0$. Hence we defi the absolute value or magnetude of a real numbe 'a' is denoted by:

$|a| = a$ if $a>0$

$|a| = 0$ if $a=0$

$|a| = -a$ if $a<0$

Note: $|a|$ is real as modulus of a or mod of 'a'.

Example: (i) $|3| = 3$, $3>0$

(ii) $|-3| = -3(-3) = 3$, $-3<0$

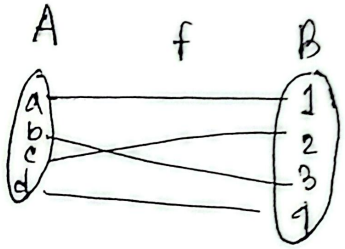
(iii) $|0| = 0$

Function : Suppose that to each element in set A is assign by some manner or other a unique element set B. We call their assignments a function from A into B and we can write $f: A \to B$. If $f: A \to B$ a function from A into B, then

(i) The set A is called domain of the function

(ii) The set B is called co-domain of the function

(iii) $f(A)$ is called the ~~the~~ range of the function Which consists of those elements of B which ap as the image of at least one elements of A where $f(A) \in B$.



Let A and B be two non empty sets, then a rule 'f' which associations each element of A with a unique elements of B is called a function from A into B. If is a function from A into B we write $f: A \to B$ (read as f is a function from A into B.



If f associates $x \in A$ to $y \in B$, then we can say that y is the image of element x under function f. It is denoted by $f(x)$ and we write $y = f(x)$

Monotone function : Let x_1, x_2 be any two points such that $x_1 < x_2$ in the interval of defination of a function $f(x)$. Then $f(x)$ is said to be monotically increasing if $f(x_1) \leq f(x_2)$ and monotically decreas if $f(x_1) \geq f(x_2)$. Thus in $[0, \pi/2]$, $\sin x$ is a exam of monotically increasing function and $\cos x$ i the example of monotically decreasing function.

Even function : Let $f(x)$ be a function defined a domain $D \in R$. The function $f(x)$ is said to be a even function if $f(-x) = f(x)$ for all $x \in$

Odd function : A function $f(x)$ issaid to be a Odd function if $f(-x) = -f(x)$ for all $x \in D$

Example: $f(x) = \cos x$

$f(-x) = \cos(-x) = \cos x = f(x)$

$\therefore f(-x) = f(x)$

so, $f(x) = \cos x$ is a even function

And $f(x) = \sin x$

$f(-x) = -\sin x = -f(x)$

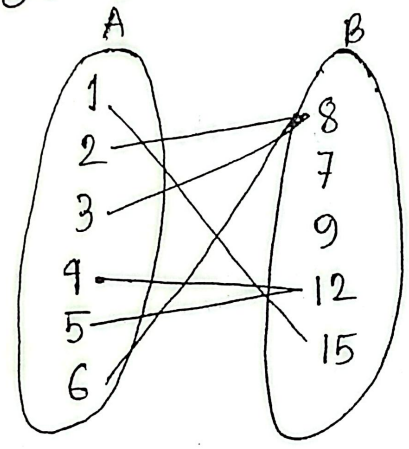
$\therefore f(-x) = -f(x)$

so, $f(x) = -\sin x$ is an odd function

Domain : Let $y = f(x)$ be a function. Then the set of all possible values of 'x' is called domain of the function.

Range : Let $y = f(x)$ be a function. then the set of all image point of 'x' is called the range of the function.

Domain, Range and co-domain in the function



The set A is called the domain of the function f and the set B is called the co-domain. The set of all elements image of A under the function f is called the range of f and is denoted by D_f and R_f.

Here, Domain of f, $D_f = \{1, 2, 3, 9, 5, 6\}$

Range of f, $R_f = \{8, 12, 15\}$

co domain of f $= \{8, 7, 9, 12, 15\}$

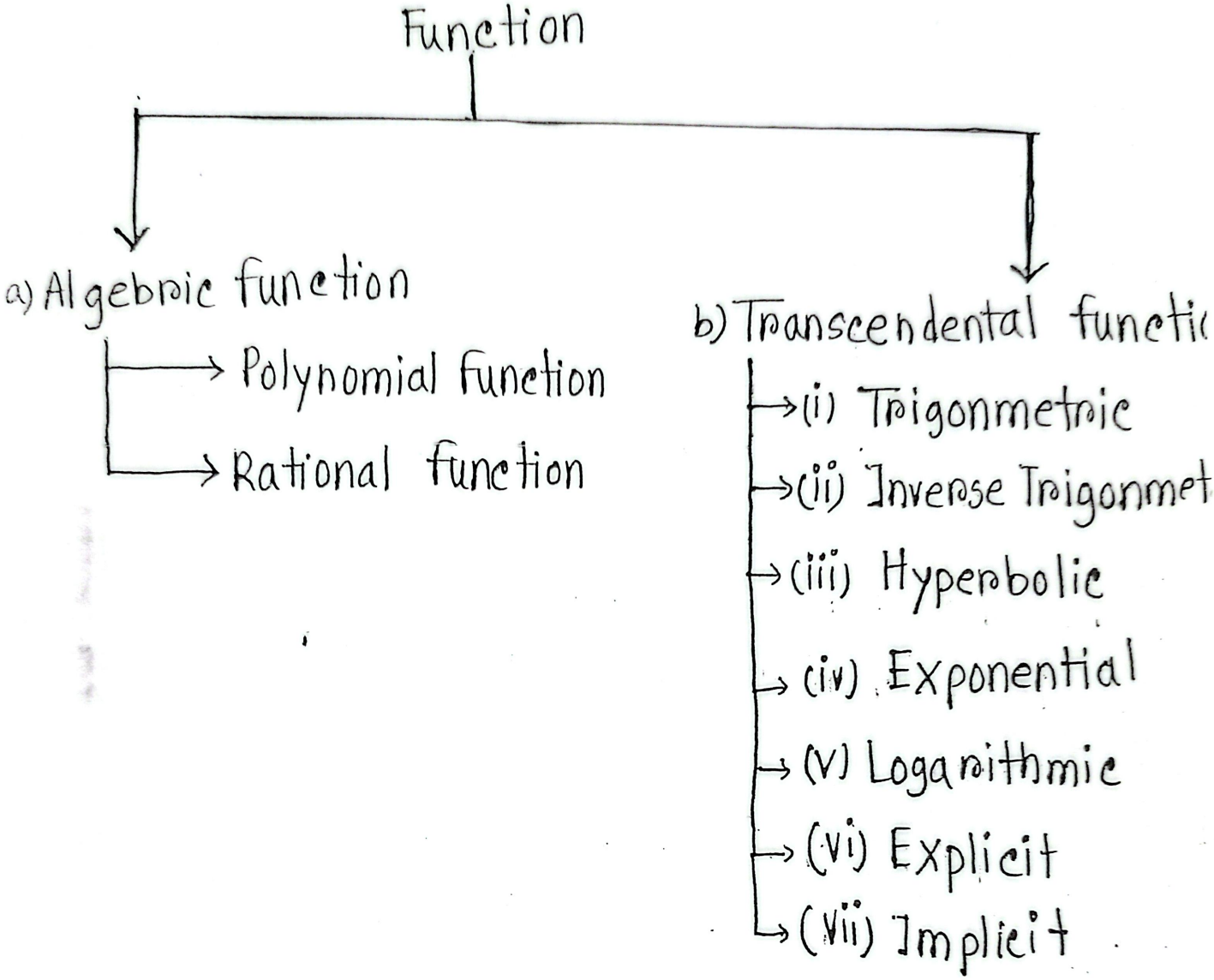
Ex: Let $f: R \to R$ be defined by $y = f(x) = \frac{1}{x}$ domain x

of f $D_f = \{x : x \in R, x \neq 0\}$

Range of f, $R_f = \{x : x \in R, y \neq 0\}$

co-domain $= \{y : y \in R\}$

#Classification of function



a) Algebraic function: Function consisting of finite numbe of terms involving power and roots of the independe variable and the four fundamental operations (+, −, ×, are called Algebraic function.

Example: (i) $3x^4 - 5x + 7$

(ii) $x^{3/2} + 5x$

(iii) $\frac{\sqrt{x+1}}{x-1}$

(i) Polynomial function : A function of the form

$$f(x) = P(x) = a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n$$

Where $a_0, a_1, a_2 \dots a_n$ are given real numbers and $a_0 \neq 0$ is called a polynomial function of degree n.

Example: (i) $P(x) = x^7 + 3x^3 + 1$ (a polynomial of degree 7)

(ii) $P(x) = x^3 - 3x + 1$ (a polynomial of degree 3)

(iii) $P(x) = 3x^0$ (a polynomial of degree 0)

(ii) Rational function : A rational function is the ratio of the polynomials of which the denomination is not identically zero.

Thus if $P(x)$ and $Q(x)$ be two functions and $Q(x)$ is not identically zero. Then $R(x) = \frac{P(x)}{Q(x)}$ will be a rational function of x.

Note : If we choose $Q(x) = 1$, We get $R(x) = P(x)$ showing that every polynomial is a rational function.

Example: (i) $\frac{7x^4 - x^2 + 2}{x^2 - 4x + 3}$ is a rational function

Which is defined for all real values of x exepts 1 and 3.

(ii) $x^2 + 3x + 5$ is a rational function

b) Transcendental function: A function which is non-Algebraic is called a Transcendental function.

(i) Trigonmetric function: That function is said to a trigonmetric function which involves circular functions (sine, cosine, tangent, cotangent, secant, cosecant) of variable angles.

Example: (i) $f(x) = \sin x$, $f(x) = \cos x$, $f(x) = ta$

(ii) Inverse trigonmetric function:

$y = \sin^{-1} x \quad (-\pi/2 \leq y \leq \pi/2)$

$y = \cos^{-1} x \quad (0 \leq y < \pi)$

$y = \tan^{-1} x \quad (-\pi/2 \leq y \leq \pi/2)$

(iii) Exponential function: If $a > 0$, $a \neq 1$ then t function defined by

$y = a^x \quad x \in \mathbb{R}$ is called an exponential function with base a, $y = a^x$ is defined for all x and $0 < y < \infty$

Example: (i) $y = 2^x$

(ii) $y = e^x$

Explicit function : A function is said to be an Explicit function of x if the dependent variable y can be expressed interms of independent variable x only.

(i) $y = x^2 + 2$

(ii) $y = \sin x + \cos^2 x$

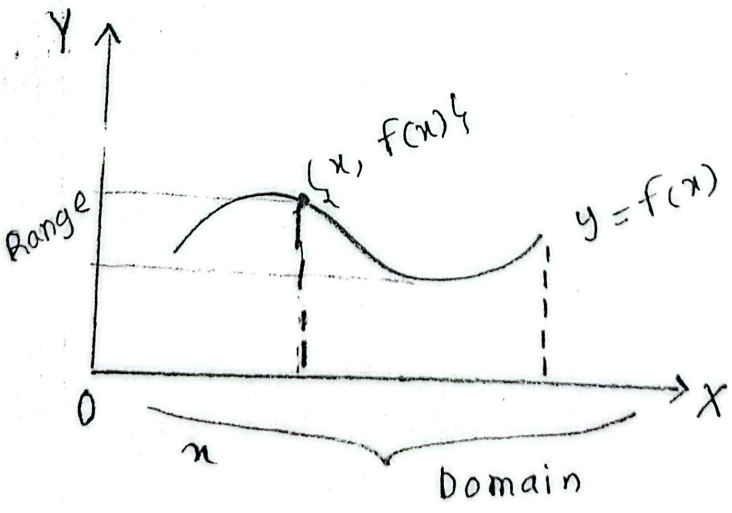
Implicit function : A function is said to be an Implicit function, when it cann't be expressed directly in terms the independent variable.

Example: (i) $f(x,y) = x^2 + y^2 - 1 = 0$

(ii) $f(x,y) = x^3 + 5x^2y + 86y^3 = 0$

Graphical Representation of a function:

$y = f(x)$



The set of points $(x, f(x))$ obtained by giving diffe values of x, is said to be the graph of the function $f(x)$. Also $y = f(x)$ is said to be the equa of the graph.

Domain and Range of a function:

The set of x values for which a function $Y = f(x)$ is well defined is called the domain of the functio and is denoted by D_f.

Note: set of the values of the independent variab

Example: find the domain of the following funct

(i) $y = f(x) = \frac{1}{x}$, $D_f = \{x : x \in R, x \neq 0\}$

(ii) $y = f(x) = \sqrt{P}$, $D_f = \{P \geq 0\}$

(iii) $y = f(x) = \frac{1}{\sqrt{Q}}$, $D_f = \{Q > 0, Q \neq 0\}$

(iv) $y = f(x) = \frac{1}{R}$, $D_f = R - \{0\}$

(v) $y = f(x) = \log T$, $D_f = \{T > 0\}$

Range: The set of values of Y, that are associate with the set of values of x in the domain of t

function $y = f(x)$ is called the range of the function and is denoted by R_f

Note : set of values of dependent variable

Example: find the domain and range of the following function

(i) $y = f(x) = \sqrt{x^2 - 4}$, $D_f = \{x : x \leq -2 \; ; \; x \geq 2\}$

$R_f = \{y : y \geq 0\}$

(ii) $y = \frac{2}{x-3}$,

$D_f = \{x : x \in \mathbb{R}, x \neq 3\}$

$R_f = \{y : y \in \mathbb{R} \; ; \; x \neq 0\}$

Example 01:

Draw the graph of the following function

$$f(x) = \begin{cases} x & \text{when } x \geq 0 \\ -x & \text{when } x < 0 \end{cases} \quad \text{or } f(x) = |x|$$

Find the domain and range of the above function

Sol^n: We have,

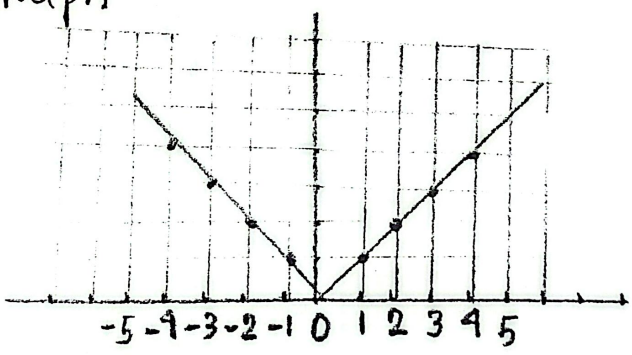
$$y = f(x) = \begin{cases} x & \text{when } x \geq 0 \\ -x & \text{when } x < 0 \end{cases}$$

For the graph of the functions, we have the follov tables

x	0	1	2	3
$y=f(x)$	0	1	2	3

x	-1	-2	-3	-4
$f(x)=y$	1	2	3	4

By the help of the values of the tables, we have following graph



Domain of $f(x) = [0, \infty) \cup (-\infty, 0]$

$= (\infty, -\infty) = \{x : x \in \mathbb{R}\}$

Range of $f(x) = [0, \infty) = \{y : y \in \mathbb{R}^+ \text{ with zero}\}$

Example 02:

Draw the graph of the following function

$$f(x) = x^2$$

Find the domain and range of the above function

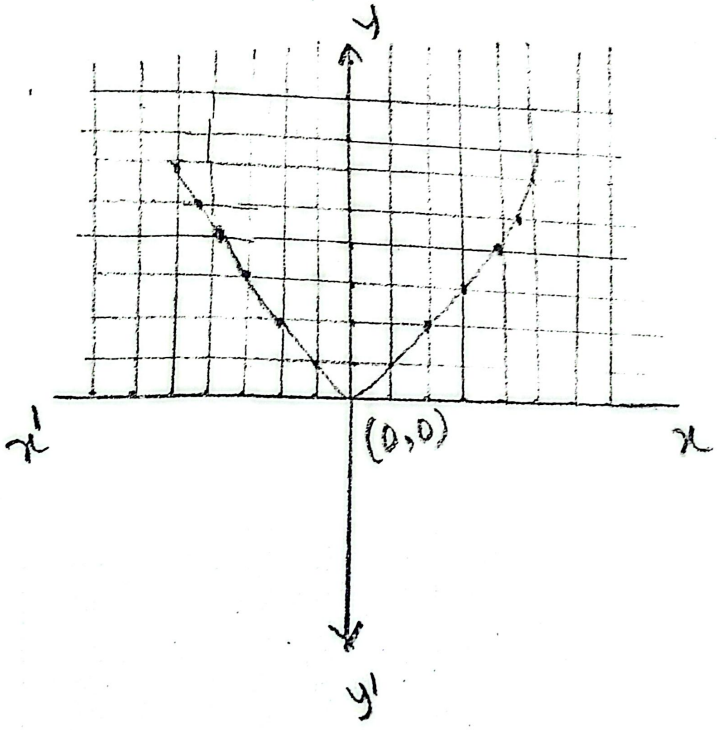
Soln: We have,

$$f(x) = x^2$$

For the graph of the following function, we have tables,

x	0	1	2	3	4
$y=f(x)$	0	1	4	9	16

x	-1	-2	-3	-4
y	1	4	9	16



Domain of $f(x) = \{x : x \in \mathbb{R}\}$

Range of $f(x) = \{y : y \in \mathbb{R}^+ \text{ with zero}\}$

$\therefore [0, \infty)$

Example 03:

Draw the graph of the following function $f(x) = \sqrt{x}$ find the domain and range of the function.

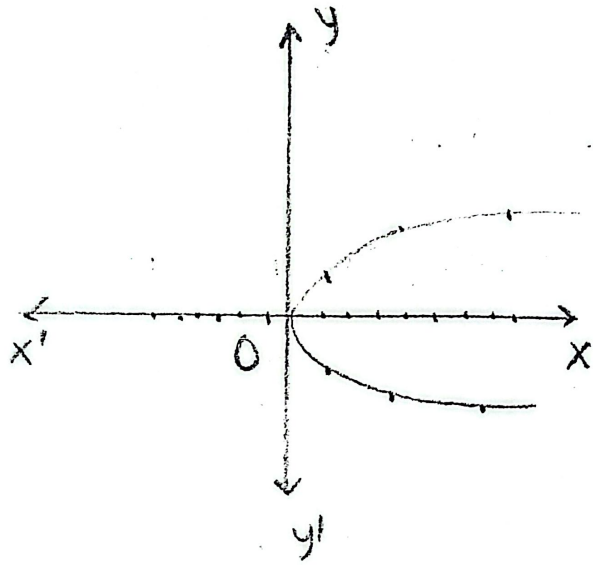
Soln: We have,

$$y = f(x) = \sqrt{x}$$

For the graph of the function, we have the following table

x	0	1	4	9
$y = f(x)$	0	± 1	± 2	± 3

By the help of the values of the table, we have the following graph



Domain of $f(x)$, $D_x = [0, \infty)$ or $\{x : x \in \mathbb{R}^+$ with zero

Range of $f(x)$, $R_x = (-\infty, \infty)$ or $\{y : y \in \mathbb{R}\}$

Example 01:

Draw the graph of the following function

$$f(x) = \begin{cases} -1 & \text{when } x \leq -1 \\ x & \text{when } -1 < x < 1 \\ 1 & \text{when } x \geq 1 \end{cases}$$

find the following function's domain and range.

Soln: We have,

$$f(x) = \begin{cases} -1 & \text{when } x \leq -1 \\ x & \text{when } -1 < x < 1 \\ 1 & \text{when } x \geq 1 \end{cases}$$

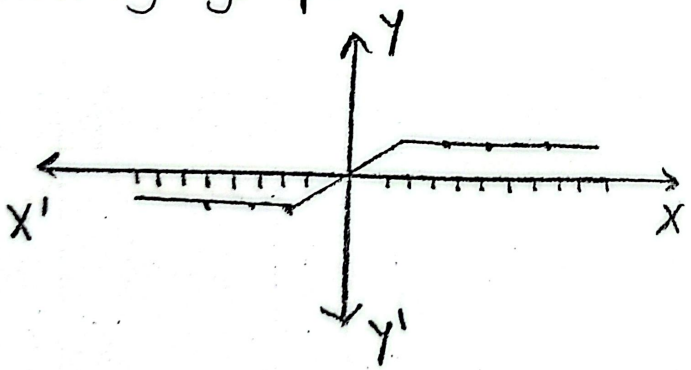
for the graph of the function, we have the following tables

x	-1	-2	-3	-4	-5
$y=f(x)$	-1	-1	-1	-1	-1

x	0	0·5	-0·5
$y=f(x)$	0	0·5	-0·5

x	1	2	3	4	5
y	1	1	1	1	1

By the help of the values of the table, we have the following graph



Domain of $f(x) = (-\infty, \infty)$

Range of $f(x) = [-1, +1]$

Example 05:

Draw the graph of the function defined by $f(x) = \frac{1}{2}(|$ find the domain and range of the above function.

Soln:

We have,

$$y = f(x) = \frac{1}{2}(|x| + x)$$

$$= \frac{1}{2}(x+x) \text{ where } x > 0$$

$$= \frac{1}{2}(-x+x) \text{ where } x < 0$$

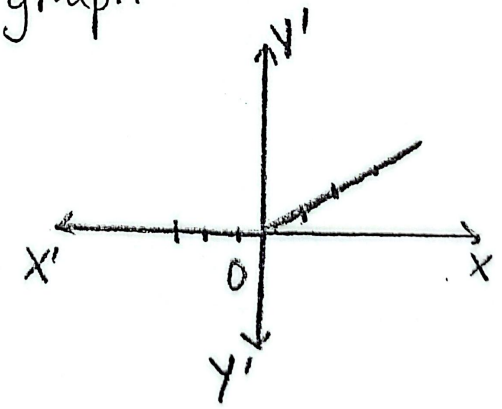
$$= \frac{1}{2}(0+0) \text{ where } x = 0$$

for the graph of the function, we have the following tables

x	1	2	3	4
y	1	2	3	4

x	-1	-2	-3	0
y	0	0	0	0

By the help of the values of the table we have the following graph



Domain of $f(x) = (-\infty, \infty)$

Range of $f(x) = [0, \infty)$

Example 06:

We have,

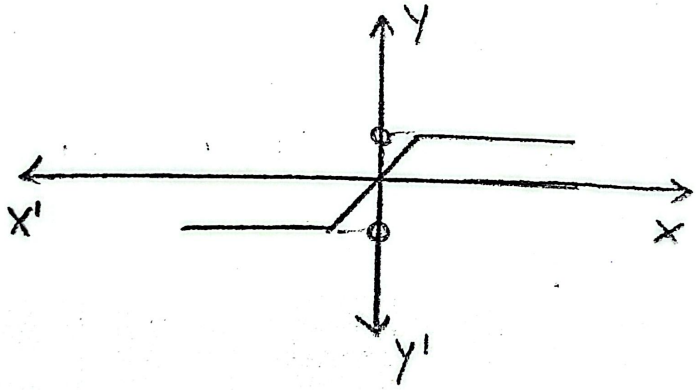
$$y = f(x) = \begin{cases} \frac{|x|}{x}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$$

$$\Rightarrow y = f(x) = \begin{cases} \frac{x}{x} & \text{when } x > 0 \\ \frac{-x}{x} & \text{when } x < 0 \\ 0 & \text{when } x = 0 \end{cases}$$

x	1	2	3	4	5	0
$y=f(x)$	1	1	1	1	1	0

x	-1	-2	-3	-4
$y=f(x)$	-1	-1	-1	-1

By the ~~graph~~help of the values from table, we have the following graph



Domain $f(x) = (-\infty, \infty)$

Range $f(x) = [1, 0, -1]$

Example 07:

Draw the graph of the following function,

$$f(x) = |x| - x$$

find the domain and range of the above functio

Soln: We have,

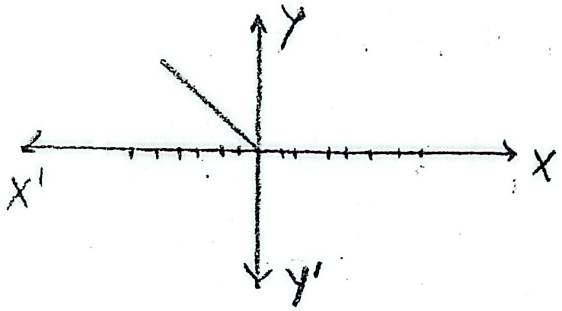
$$y = f(x) = |x| - x$$

$$= \begin{cases} x - x, & \text{when } x > 0 \\ -x - x, & \text{when } x < 0 \\ 0, & \text{when } x = 0 \end{cases}$$

for the graph of the function:

x	1	2	3	4	0
$y = f(x)$	0	0	0	0	0

x	-1	-2	-3	-4
$y = f(x)$	2	4	6	8

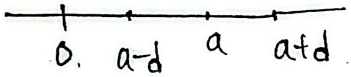


Domain of $f(x) = (-\infty, \infty) = \{x : x \in \mathbb{R}\}$

Range of $f(x) = [0, \infty) = \{y : y \in \mathbb{R} \text{ with } 0\}$

Limit and continiousty of function

$\oint$ A set of points lying between any two points 'a' and 'b' is said to be an interval. The closed interval includes the end points 'a' and 'b' also and is denoted by $[a, b]$ which the open interval does not include the points and is denoted by (a, b). A neighbourhood of a point 'a' is an open interval containing the point a, it is generally denoted by $(a-\delta, a+\delta)$



which is obviously an open interval containing 'a'. But we find it more convenient to use the equivalent absolute value from in into place.

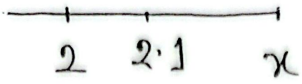
Let $x \leftarrow (a-\delta, a+\delta)$ Then

$$a-\delta < x < a+\delta$$

$$\Rightarrow -\delta < x-a < \delta$$

$|x-a| < \delta$ Hence $x \leftarrow (a-\delta, a+\delta)$ is equavalent to $|x-a| < \delta$

§ Cocept of limit :

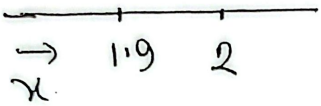


$x = 2{\cdot}1, 2{\cdot}01, 2{\cdot}001, 2{\cdot}0001, 2{\cdot}00001 \ldots$

$|x-2| = 0{\cdot}1, 0{\cdot}01, 0{\cdot}001, 0{\cdot}0001, 0{\cdot}00001 \ldots$

(δ Delta)

δ is the limit of 2·1



$x = 1{\cdot}9, 1{\cdot}99, 1{\cdot}999, 1{\cdot}9999 \ldots$

$|x-2| = 0{\cdot}1, 0{\cdot}01, 0{\cdot}001, 0{\cdot}0001 \ldots$

We consider the value of $f(x)$ when $x \neq 2$, but t
nearest to 2 and $x > 2$

x	2·1	2·01	2·001
$f(x) = y = x+2$	4·1	4·01	4·001

it is clean from the above table not x approches
that is $x \to 2$.

Through the values greater than 2, $f(x)$ approches
that is $f(x) \to 4$ we will explain this fact by say
that right hand limit of $f(x)$ as $x \to 2$ exists
is equal to 4 and in symbols we will write lim

$f(x) = 0$ or $\lim f(x) = 4$